

Ig-like domain-containing protein of *Wuchereria bancrofti*: a therapeutic target for a multi-epitope subunit vaccine against lymphatic filariasis

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Introduction

Lymphatic filariasis is a neglected tropical disease that is endemic in tropical regions and affects 120 million people worldwide. It is caused by filarial worms mainly *Wuchereria bancrofti*.



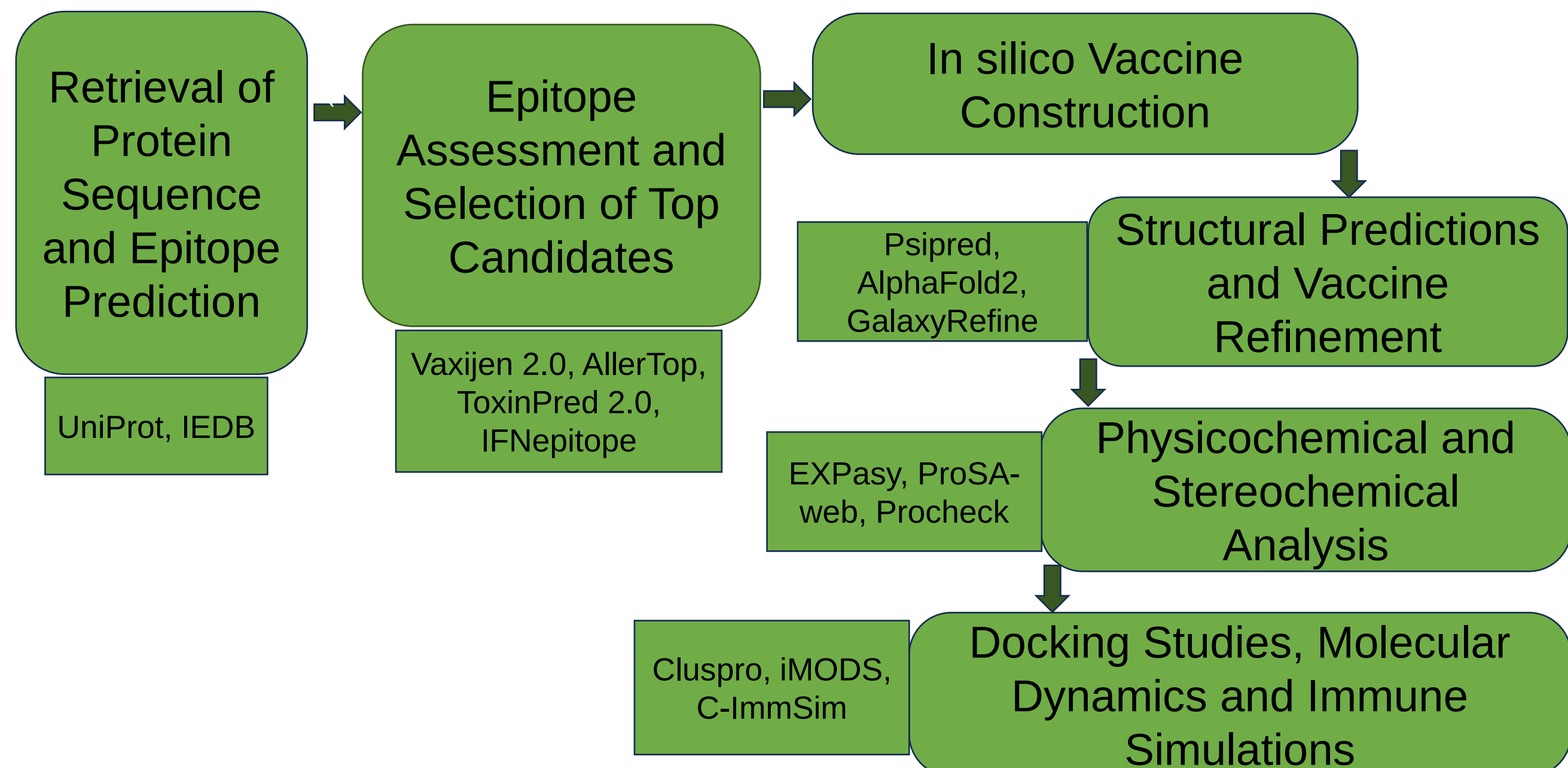
Figure 1: Clinical manifestation of LF

The disease culminates in lymphedema shown in Figure 1 which inflicts severe pain on the patients. While efforts have been made by WHO to eliminate the disease, existing drugs for the treatment of LF have been reported to be ineffective due to resistance¹.

Objectives

- To construct a vaccine using immunological epitopes
- To analyze the structural properties of the vaccine
- To simulate the immune response to the vaccine construct

Methods



Results and Discussion

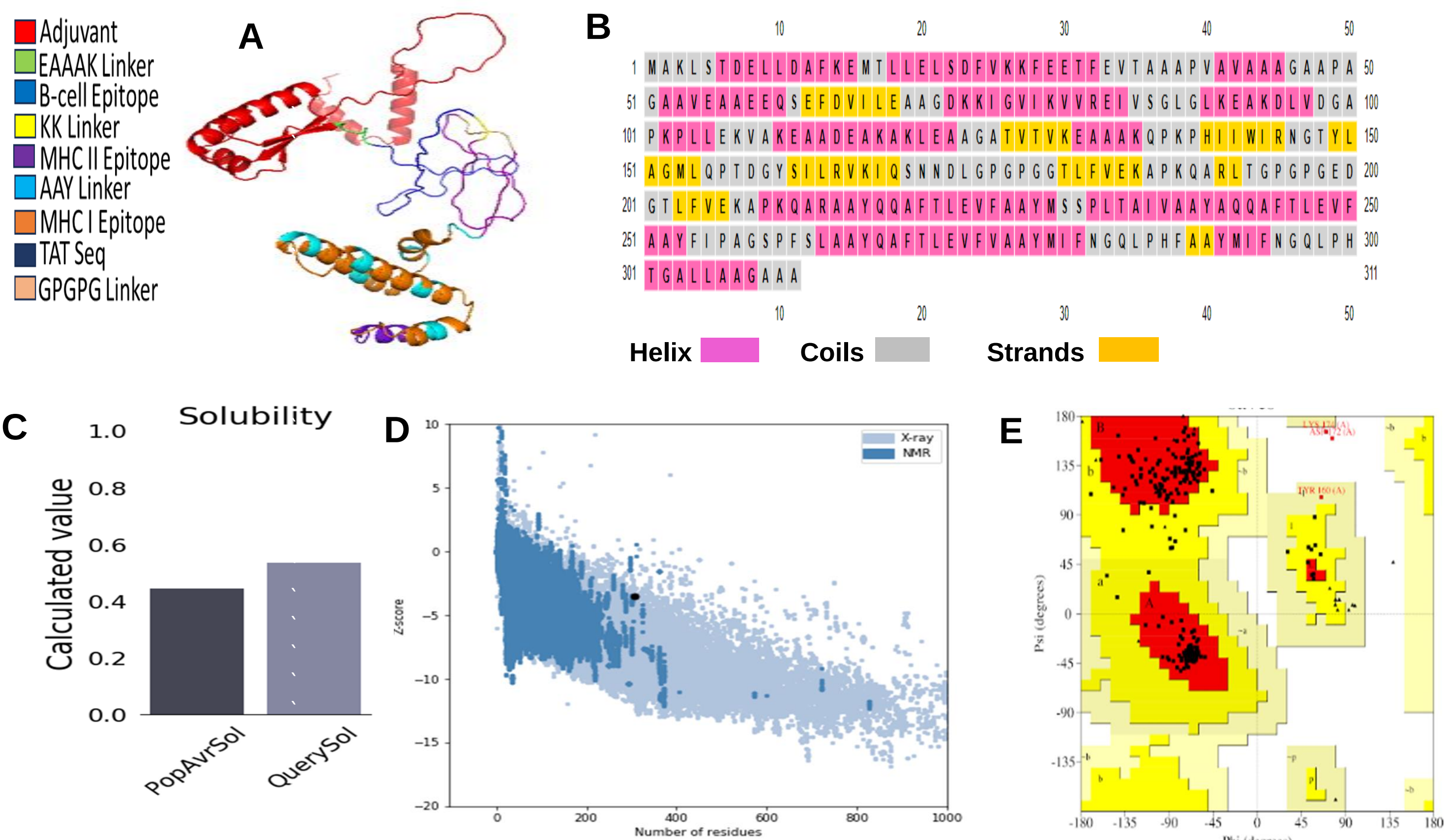


Figure 2: Structural analysis of the multi-epitope vaccine construct. (A) 3D model of the vaccine, (B) Secondary structure, (C) Solubility, (D) Z-score plot, (E) Ramachandran plot. The structural analysis demonstrates the candidate vaccine is stable, soluble, and has desirable quality compared with other peptides or proteins.

Table 1: Physicochemical properties of the vaccine construct

Physicochemical properties	Score	Remark
Aliphatic index	94.03	Thermally stable
Instability index	32.86	Stable
Number of amino acids	308	-
Theoretical pI	5.30	Acidic
Solubility	0.538	Soluble
Molecular weight	32559.7	-
Chemical formula	C ₁₄₈₉ H ₂₃₄₂ N ₃₇₂ O ₄₃₂ S ₆	-
Estimated half-life(mammalian reticulocytes, in vitro)	30 hours	-
Estimated half-life(yeast cells, in vitro)	> 20 hours	-
Estimated half-life(E-coli, in vitro)	> 10 hours	-

Results and Discussion

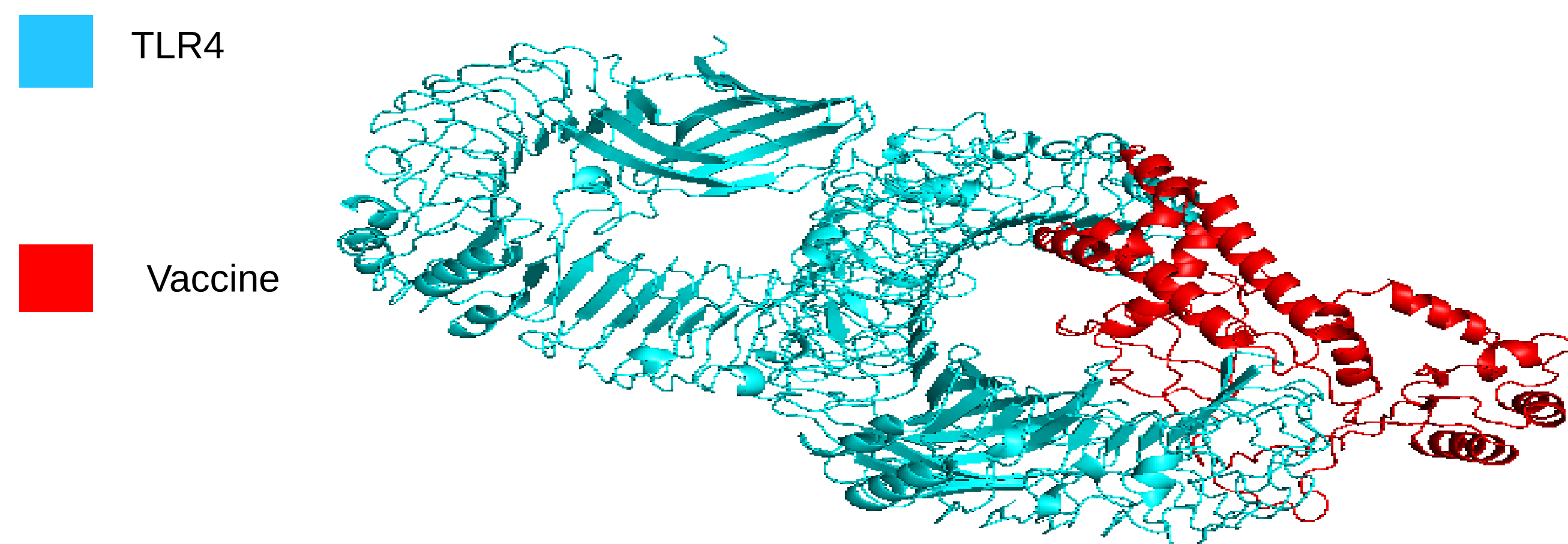


Figure 3: The docked complex of the candidate vaccine peptide and the TLR4. Molecular docking analyses indicate the candidate vaccine's ability to interact effectively with TLR4

Table 2: Vaccine –TLR4 complexes

Complex	Members	Weighted score(Energy)
1	30	-1211.2
2	55	-1080.9
3	37	-1079.8

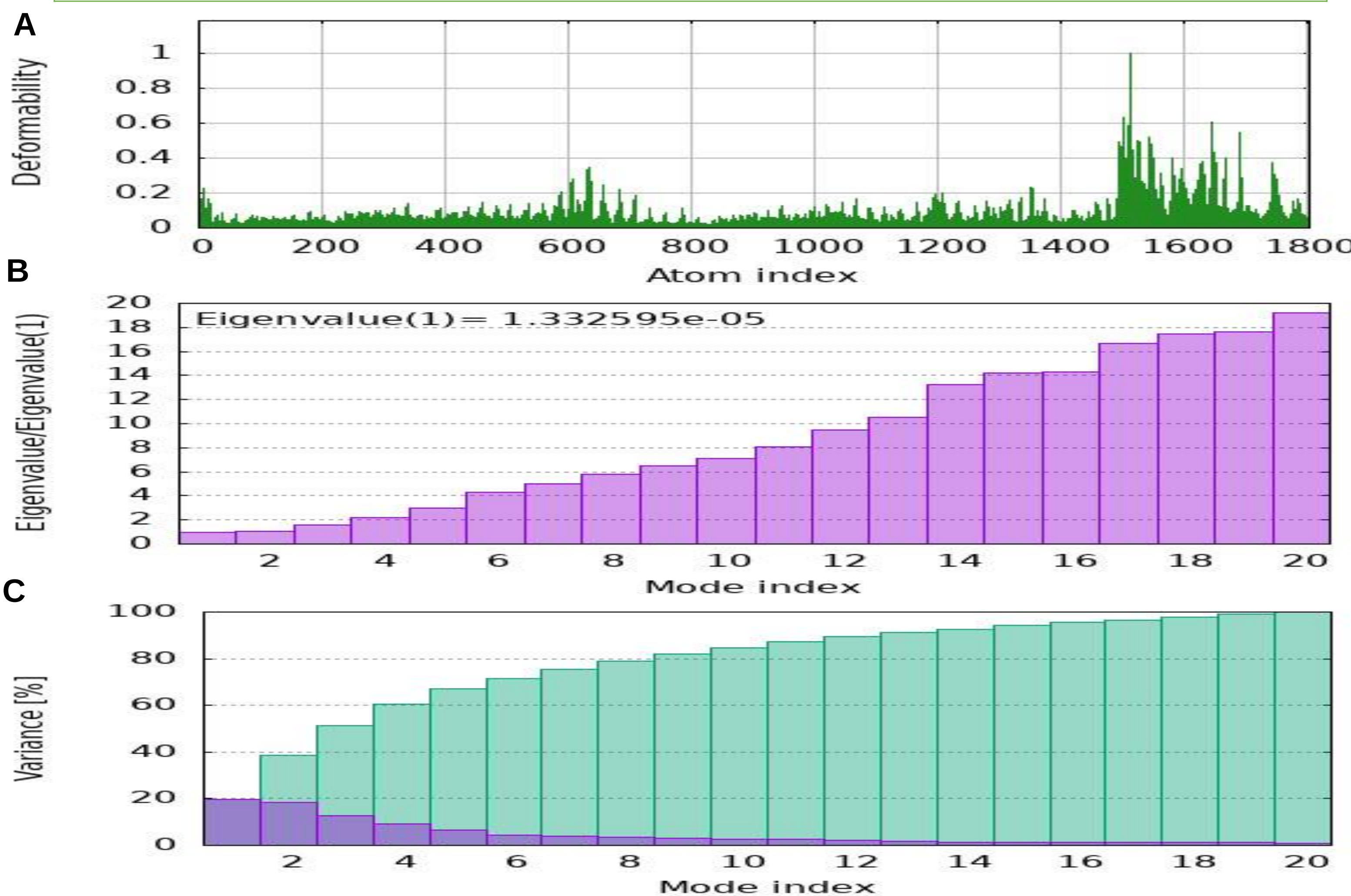
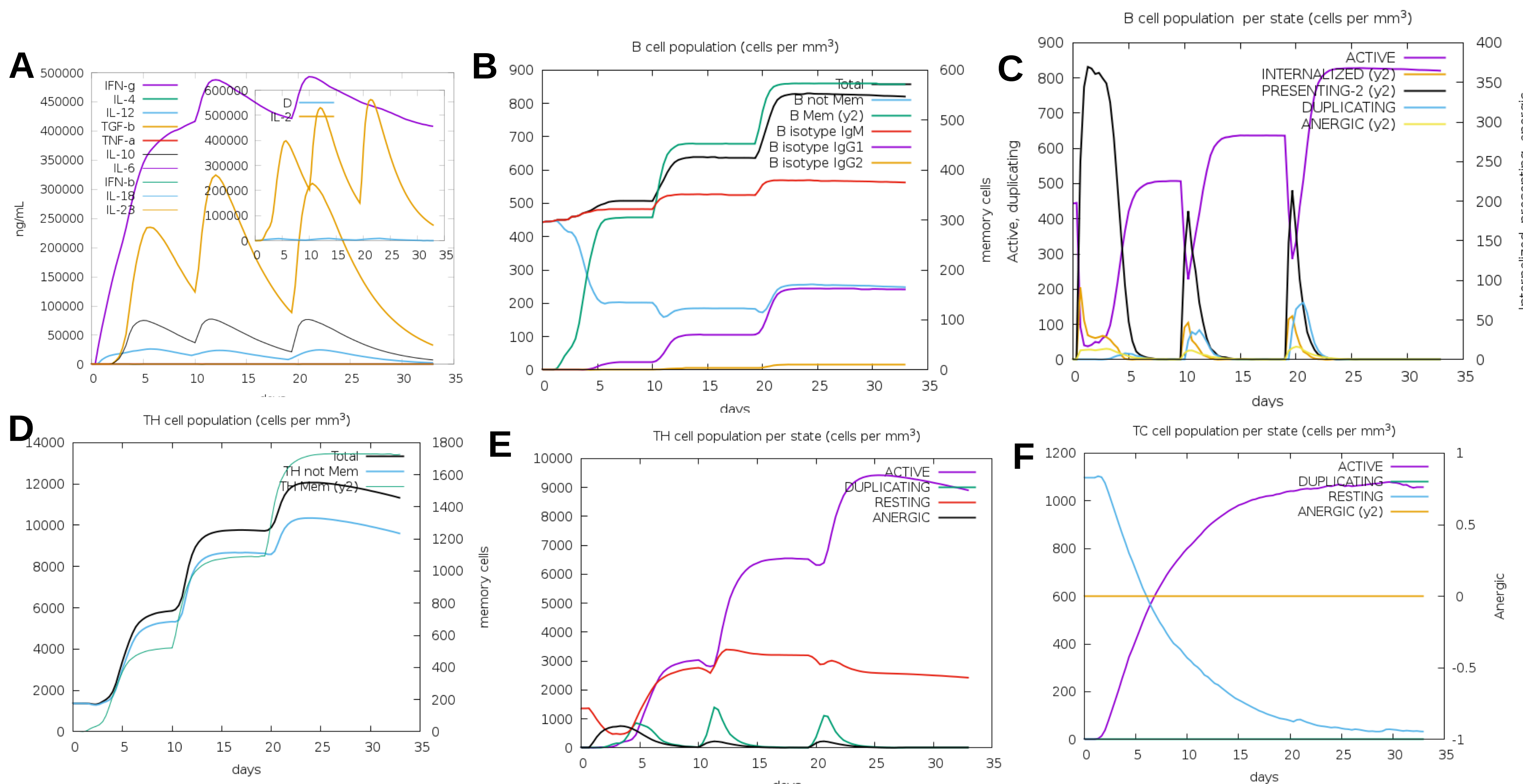


Figure 4: Molecular dynamics analysis of the vaccine candidate. (A) Deformability plot, (B) Eigenvalue plot, and (C) Variance plot. Insights from the deformability plot show flexible and rigid regions in the vaccine for interaction with immune receptors like TLR4.



Conclusion

The candidate vaccine elicited a robust immune response against the filarial antigens during the simulation and has desirable physicochemical properties. The preliminary results indicate the model vaccine can be considered for further experimentation.

References

- N. Sangshetti, J., B. Shinde, D., Kulkarni, A. & Arote, R. Two decades of antifilarial drug discovery: a review. (2017) doi:10.1039/C7RA01857F.
- Kwarteng, A., Asiedu, E., Koranteng, K. K. & Asiedu, S. O. Highlighting the Relevance of CD8+ T Cells in Filarial Infections. *Front. Immunol.* **12**, 714052 (2021).